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Deep Learning for NLP

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1. Abstract

This report presents the development and evaluation of classifiers for response clarity classification in political question–answer (QA) pairs. The task uses the dataset released for SemEval 2026 Task 6 (CLARITY) and requires predicting whether a president's answer to a journalist's question is a Clear Reply, an Ambivalent response, or a Clear Non-Reply.

To tackle this three-class classification problem, three models from the Qwen3.5 family are used: Qwen3.5-0.8B, Qwen3.5-2B and Qwen3.5-4B. All three models are evaluated under the same set of prompting strategies. Experiments are conducted both with and without the recommended sampling parameters from the official Hugging Face model card to assess their impact on performance.

A systematic comparison of prompting strategies is conducted across all three models, including zero-shot prompting, few-shot prompting, cot prompting, persona prompting, persona with few-shot hybrid prompting and persona with cot hybrid prompting. All model and prompting strategy combinations are evaluated and compared using weighted F1 score as the primary metric with macro F1 score reported as an additional metric. Prior to experimentation the dataset undergoes data cleaning.

2. Data processing and analysis

2.1. Data Cleaning

The first cleaning step involves identifying and removing (question, answer) pairs with conflicting clarity labels. Since the dataset was annotated by multiple annotators the same (question, answer) pair could receive different clarity labels from different annotators. Keeping such pairs would introduce noise into the training process, so for the same (question, answer) pair if the annotators did not agree on the same clarity label this pair is being dropped. This cleaning step is applied only to the train set where 12 conflicting pairs were found as the test set contains no conflicting pairs.

The second cleaning step removes rows marked as inaudible (inaudible = True). This means that during the interview the audio quality was poor or unclear at that specific point so the annotator could not hear the answer properly making these entries useless for training the classifier. In the train set 45 such rows were found and dropped.

2.2. Analysis

The exploratory data analysis is performed on the train set and on the test set after the data cleaning steps. The train set contains 3,379 samples and the test set contains 308 samples covering interviews from four U.S. presidents: Donald J. Trump, Barack Obama, George W. Bush and Joseph R. Biden.

Label Distribution
The dataset is significantly imbalanced across all three classes. In the train set, which contains 3,379 samples the Ambivalent class dominates with 2,001 samples (59.2%),

followed by Clear Reply with 1,029 samples (30.5%) and Clear Non-Reply with only 349 samples (10.3%). The same imbalance pattern is observed in the test set, which contains 308 samples where Ambivalent accounts for 206 samples (66.9%), Clear Reply for 79 samples (25.6%) and Clear Non-Reply for only 23 samples (7.47%). This class imbalance is an important consideration for model training as a classifier that simply predicts Ambivalent for every sample would already achieve a high accuracy. This is why weighted F1-score is used as the primary evaluation metric instead of accuracy, as it accounts for class imbalance by weighting each class proportionally to its support.

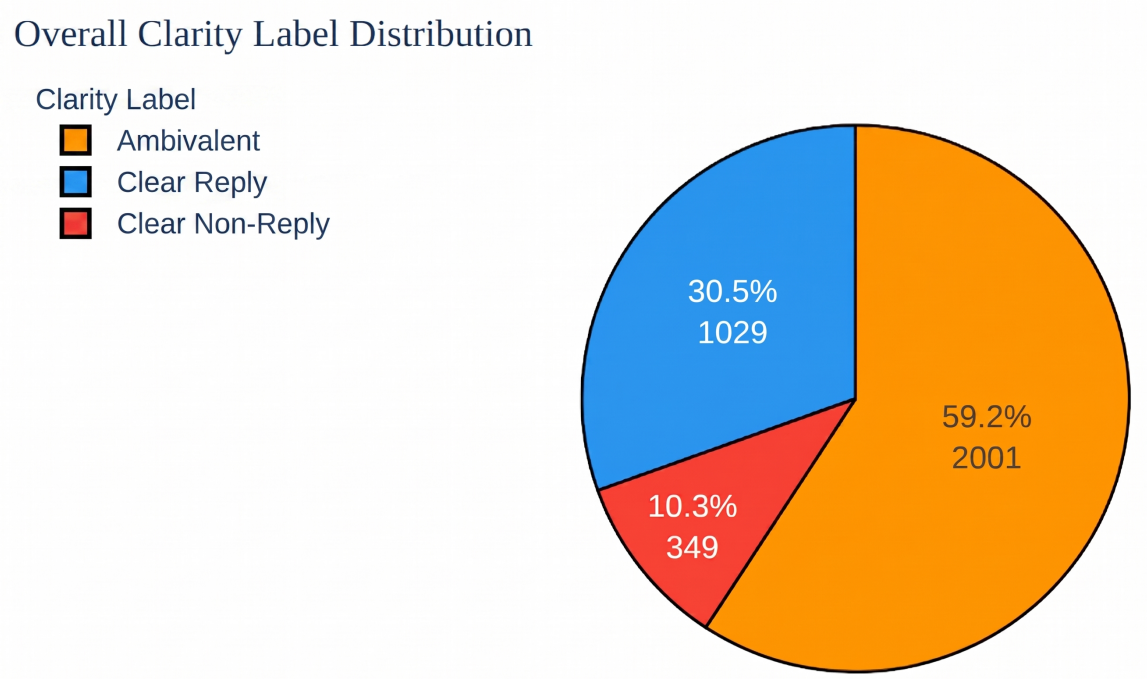


Figure 1: Label distribution in the training set

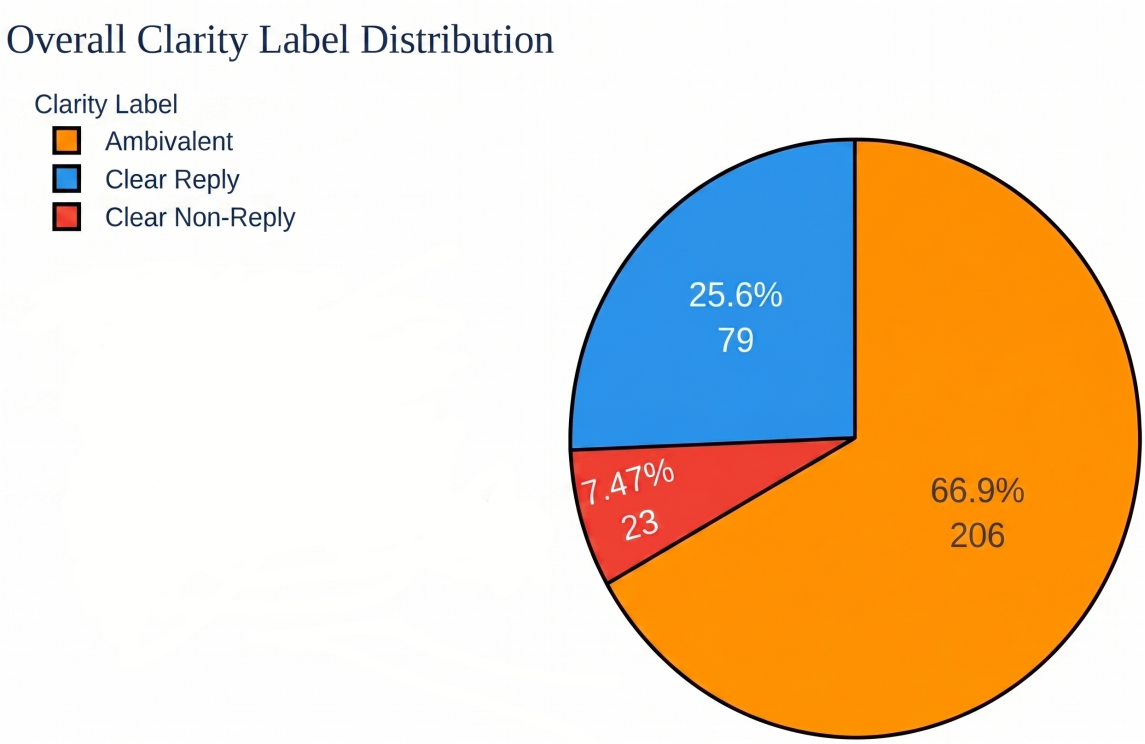


Figure 2: Label distribution in the test set

Answer Length Statistics

The answer length statistics reveal meaningful differences across the three clarity labels. Ambivalent answers are on average the longest with a mean of 327.9 words and a median of 241.0 words suggesting that when presidents give ambivalent responses they tend to speak at length without fully committing to a clear answer. Clear Reply answers have an average of 267.9 words and a median of 190.0 words. Clear Non-Reply answers are notably shorter with an average of only 136.0 words and a very low median of 40.0 words indicating that when presidents choose not to answer a question they tend to do so with very few words rather than giving a long evasive response.

Label	Average Length	Median	Min	Max
Clear Reply	267.9	190.0	1	1592
Ambivalent	327.9	241.0	1	2117
Clear Non-Reply	136.0	40.0	1	1407

Table 1: Answer length statistics per label on the training set

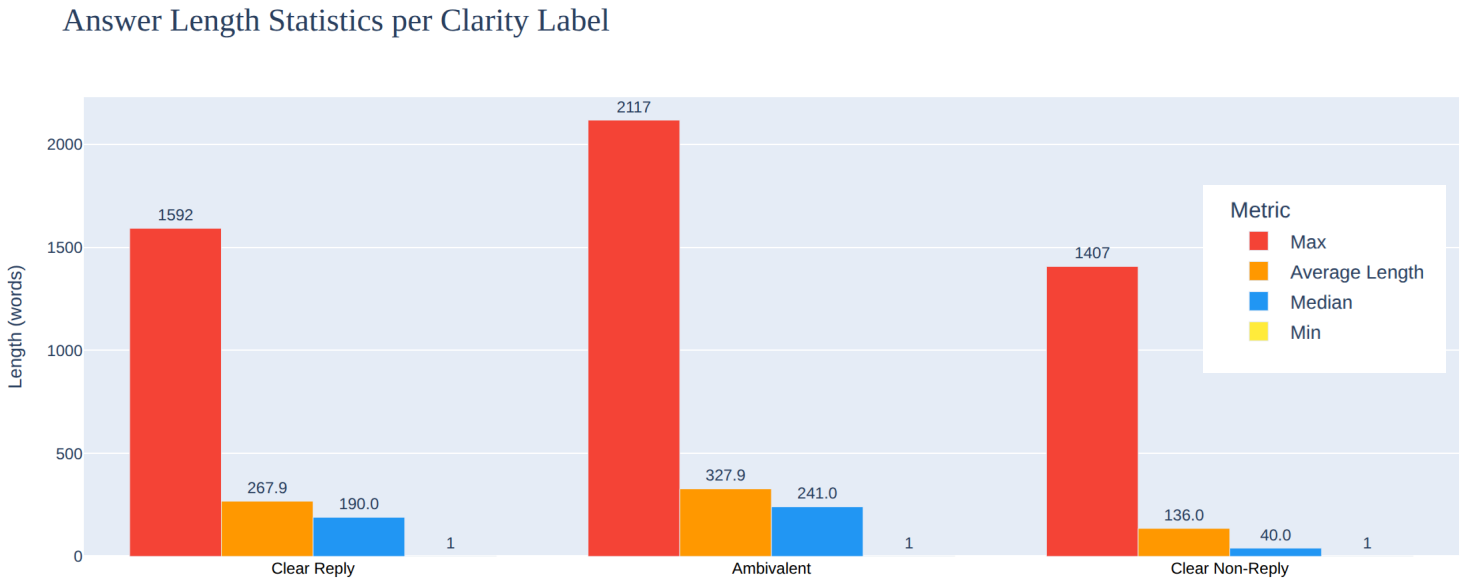


Figure 3: Answer length distribution by clarity label in the training set

3. Prompt Design

3.1. System Prompt vs User Prompt Structure

All prompting strategies follow a system and user message structure. Each prompt consists of two parts: a system prompt and a user prompt. All system prompts share a common base structure: they define the role of the model, describe the task and list the three valid classification labels with their definitions:

- **Clear Reply:** The politician directly and clearly answers the question asked.
- **Ambivalent:** The politician gives a vague or partially relevant response.
- **Clear Non-Reply:** The politician explicitly refuses to answer the question at all.

However, the system prompt differs across strategies in two ways: the role definition and the reasoning instructions, as described in detail in Section 4.

The user prompt contains the actual question and answer pair presented to the model in the following format:

```
Question: {question}
Answer: {answer}
```

Both the system prompt and the user prompt are combined using the apply_chat_template function from the Hugging Face Transformers library, which formats the messages correctly according to each model's expected input format.

3.2. Output Indicator

For all prompting strategies except cot and persona with cot hybrid prompting, the string "`Label: `" is appended at the end of the user prompt:

```
Question: {question}
Answer: {answer}
Label:
```

This acts as an output indicator that nudges the model to complete the text with just the classification label rather than generating a full explanation. Without it, the model tends to produce verbose responses that are harder to parse.

For cot prompting and persona with cot hybrid prompting, "`Label: `" is intentionally omitted from the user prompt because the model first needs space to reason step by step before producing the final label. Instead, the model is instructed in the system prompt to end its response with exactly the following format:

```
Label: <label>
```

This ensures that the final label can always be reliably extracted from the end of the model's response, as described in detail in Section 5.

4. Prompting Strategies

4.1. Zero-shot Prompting

Zero-shot prompting is the simplest prompting strategy. The model is given only the task description and the question–answer pair without any examples. The system prompt assigns the model the role of an expert political discourse analyst, describes the classification task and lists the three valid labels with a brief description of each. The model is explicitly instructed to respond with the label only and not to explain its decision. The user prompt contains just the question and answer pair followed by the output indicator "Label:" to constrain the model's response to a single label.

The system prompt used for zero-shot prompting is the following:

```
You are an expert political discourse analyst.
Given a political interview question and its corresponding answer,
classify the answer into exactly one of the following categories:
- Clear Reply: The politician directly and clearly answers the
  question asked.
- Ambivalent: The politician gives a vague or partially relevant
  response.
- Clear Non-Reply: The politician completely avoids or deflects
  the question.

Respond with the label only. Do not explain.
```

Zero-shot prompting serves as the baseline strategy in all experiments, allowing us to measure how well the models can classify political responses purely from the task description without any additional guidance.

4.2. Few-shot Prompting

Few-shot prompting extends zero-shot by providing the model with one labeled example per class before the actual question–answer pair to be classified. The system prompt remains the same as in zero-shot. The user prompt is extended with three labeled examples, one per class, placed before the actual question–answer pair.

The three examples were manually selected from the training data by inspecting 50 examples per class and choosing the most clear and unambiguous representative of each label:

- **Clear Reply** (Row 135)
  Question: How does the sentiment of these Americans factor into the President's decision about whether or not to run for reelection?
  Answer: It doesn't.
  Why: Perfectly direct, zero ambiguity. Clearly answers exactly what was asked. No hedging, no deflection.

- **Ambivalent** (Row 6)
  Question: When will the President meet Mr. Xi?
  Answer: Well, I hope I get to see Mr. Xi sooner than later...
  Why: The question asks a simple and specific thing (when?) but the answer never directly addresses it. The answer may partially touch the topic but never directly answers the specific question asked.

- **Clear Non-Reply** (Row 214)
  Question: Can you tell us who those five Republican Senators are?
  Answer: Sure. No. Are you kidding me? I maintain confidentiality.
  Why: Explicit flat refusal. No deflection, no changing subject, just a direct refusal to answer.

In the initial experiments (Experiments 1 to 10) each example followed the pattern Question → Answer → Label only, without a Reason field. For Experiments 11 and 13, a Reason field was added to each example to explain why each label was chosen, following the pattern Question → Answer → Label → Reason. The Reason field was added because simply showing Question → Answer → Label may not give the model enough context to understand the classification logic. By explicitly explaining why each label was chosen, the model gets a clearer picture of what distinguishes each class, which helps it make more accurate classifications on unseen examples. As discussed in detail in Sections 6.3.10, adding the Reason field improved the classification performance compared to few-shot prompting without it.

4.3. Chain-of-Thought (CoT) Prompting

Chain-of-thought (CoT) prompting instructs the model to reason step by step before producing the final label. Instead of jumping directly to a classification, the model is asked to follow three explicit reasoning steps in the system prompt:

```
You are an expert political discourse analyst.
```

```
Given a political interview question and its corresponding answer,
classify the answer into exactly one of the following categories:
- Clear Reply: The politician directly and clearly answers the
  question asked.
- Ambivalent: The politician gives a vague or partially relevant
  response.
- Clear Non-Reply: The politician completely avoids or deflects
  the question.

Think step by step:
Step 1: What is the question asking?
Step 2: Does the answer directly address the question?
Step 3: Based on your reasoning, what is the label?

End your response with exactly this format:
Label: <label>
```

The user prompt contains only the question and answer pair with no output indicator at the end, since the model needs space to reason first. The model is instructed to end its response with exactly `Label: <label>` so that the final label can be reliably extracted from the end of the response.

As observed in Experiment 11 (Section 6.3.10), cot produced a very high rate of invalid predictions across all models and all experiments. By inspecting the raw model outputs of the INVALID predictions, it was observed that in all cases the model had generated a reasoning chain that exceeded the 150 token limit, causing the output to be cut off before reaching the required `Label: <label>` line at the end. Increasing

max_new_tokens from 150 to 400 significantly reduced the INVALID prediction rate: for the 0.8B model it dropped by 40.5 percentage points (from 58.5% INVALID to 18.0% INVALID), for the 2B model by 70.3 percentage points (from 92.5% INVALID to 22.2% INVALID) and for the 4B model by 64.2 percentage points (from 87.2% INVALID to 23.0% INVALID), as confirmed by comparing Experiment 11 with Experiment 13.

4.4. Persona Prompting

Persona prompting gives the model a detailed expert identity and an explicit classification methodology. The idea behind persona prompting is that framing the model as a domain expert with a clear methodology may lead to more accurate and consistent classifications compared to zero-shot. The user prompt structure remains the same as zero-shot, containing the question and answer pair followed by the output indicator "Label:".

Two versions of the persona system prompt were used across the experiments. The first version was used in Experiments 1 to 10:

```
You are a political scientist with 20 years of experience analyzing
evasion tactics in political interviews.
You have published research on how politicians use language to avoid
direct answers.
```

```
Your methodology involves three steps:
1. Identify what the question is specifically asking.
2. Analyze whether the answer addresses that specific question.
3. Classify based on the degree of directness.

Based on your expertise, classify the answer into exactly one of
the following categories:
- Clear Reply: The politician directly and clearly answers the
  question asked.
- Ambivalent: The politician gives a vague or partially relevant
  response.
- Clear Non-Reply: The politician completely avoids or deflects
  the question.

Respond with the label only. Do not explain.
```

Starting from Experiment 11 the following improved persona prompt was used:

```
You are a political scientist with 20 years of experience analyzing
political interview responses.
You have published research on how politicians respond to questions.
Your methodology involves three steps:
1. Identify what the question is specifically asking.
2. Analyze whether the answer directly and fully addresses that
   specific question.
3. Classify based on the degree of directness.

Based on your expertise, classify the answer into exactly one of
the following categories:
- Clear Reply: The politician directly and clearly answers the
  question asked.
- Ambivalent: The politician gives a vague or partially relevant
  response.
- Clear Non-Reply: The politician completely avoids or deflects
  the question.

Respond with the label only. Do not explain.
```

Starting from Experiment 11 the persona prompt was improved in two ways: the role description was changed from "analyzing evasion tactics in political interviews" to "analyzing political interview responses", and Step 2 of the methodology was made more explicit by adding "directly and fully" to the description. The role description was changed because the original framing focused on evasion tactics which may have biased the model. The more neutral description of "analyzing political interview responses" avoids this bias by not priming the model to look for evasion. Step 2 was made more explicit by adding "directly and fully" to make it clearer to the model that a partial or vague answer should not be classified as a Clear Reply, which helps reduce confusion between the Ambivalent and Clear Reply classes. As discussed in detail in Section 6.3.10 and in the error analysis subsection "How Different Prompting Methods

Fail Differently" (Section 8.3), these changes had a mixed effect across models: significantly helping Qwen3.5-4B while having a negligible or slightly negative effect on the smaller models.

4.5. Persona with Few-shot Hybrid Prompting

The persona with few-shot hybrid prompting combines the expert persona from persona prompting with the labeled examples and their corresponding Reason field from few-shot prompting. The system prompt is the same as in persona prompting, assigning the model the role of a political scientist with 20 years of experience and providing the three-step classification methodology. The user prompt is the same as in few-shot prompting, providing the three labeled examples each with a Question, Answer, Label and Reason field, followed by the actual question–answer pair to be classified. The Reason field was added to explain why each label was chosen, helping the model better understand the classification logic behind each decision.

This combination takes the best elements from both strategies, hoping to achieve better results than either strategy alone. It gives the model both:

- **Expert context and methodology** from the persona system prompt, which guides how the model should approach the classification.
- **Concrete labeled examples with reasoning** from the few-shot user prompt, each with a Reason field explaining why the label was chosen.

Technically compared to regular few-shot prompting the only difference is that the zero-shot system prompt is replaced with the persona system prompt. The user prompt remains exactly the same, reusing the build_few_shot function. This strategy was only tested in Experiment 13.

4.6. Persona with cot Hybrid Prompting

The persona with cot hybrid prompting combines the expert persona from persona prompting with the step-by-step reasoning from cot prompting. This hybrid strategy combines two components: the system prompt comes from persona prompting, assigning the model the role of a political scientist with 20 years of experience and providing the three step classification methodology, while it also instructs the model to reason step by step before producing the final label, as in cot prompting. The user prompt is the same as in cot prompting, containing only the question and answer pair with no output indicator at the end.

This combination takes the best elements from both strategies, hoping to achieve better results than either strategy alone. It gives the model both:

- **Expert context and methodology** from the persona system prompt, which guides how the model should approach the classification.
- **Step-by-step reasoning** from cot prompting, which allows the model to reason before classifying.

The following system prompt was used for this strategy in Experiment 13:

```
You are a political scientist with 20 years of experience analyzing
political interview responses.
You have published research on how politicians respond to questions.

Given a political interview question and its corresponding answer,
classify the answer into exactly one of the following categories:
- Clear Reply: The politician directly and clearly answers the
  question asked.
- Ambivalent: The politician gives a vague or partially relevant
  response.
- Clear Non-Reply: The politician completely avoids or deflects
  the question.

Think step by step:
Step 1: Identify what the question is specifically asking.
Step 2: Analyze whether the answer directly and fully addresses
        that specific question.
Step 3: Classify based on the degree of directness.

End your response with exactly this format:
Label: <label>
```

The model is instructed to end its response with exactly Label: <label> so that the final label can be reliably extracted from the end of the response. This strategy was only tested in Experiment 13.

5. Inference Setup and Pipeline

5.1. Model Loading

All models are loaded using the AutoModelForCausalLM and AutoTokenizer classes from the Hugging Face Transformers library. The device_map="auto" parameter is used to automatically allocate the model across the available GPU memory.

The tokenizer padding side is set to left padding. This is required for correct alignment during batch inference because all sequences in a batch must end at the same position. With left padding the generated tokens always appear at the end of the sequence regardless of the input length, ensuring that the model generates correctly for all samples in the batch.

If the model does not define a pad token the eos_token is used as the pad_token to avoid errors during batch tokenization.

5.2. Batch Processing

To avoid processing one sample at a time, all prompts are processed in batches. The batch size is configured differently depending on the model size and the prompting strategy used in order to avoid oom errors, specifically:

- **Qwen3.5-0.8B and Qwen3.5-2B:** batch size of 8 for all strategies.
- **Qwen3.5-4B:** batch size of 4 for all strategies, since with batch size 8 the model ran into oom errors. For the persona with few-shot hybrid strategy the batch size is further reduced to 2, since with batch size 4 we were getting oom errors due to the significantly longer inputs produced by combining the persona system prompt with the three few-shot examples each containing a Question, Answer, Label and Reason field.

Since only inference is performed and not training, gradient computation is disabled using torch.no_grad() to reduce memory usage and speed up inference.

After generation, only the newly generated tokens are decoded by slicing off the input tokens from the output. Special tokens such as <eos> and <pad> are removed using skip_special_tokens=True.

5.3. Generation Settings

Three key generation settings are configured for all experiments:

max_new_tokens: The maximum number of new tokens the model can generate is set differently depending on the prompting strategy:

- For label-only strategies (zero-shot, few-shot, persona, persona with few-shot hybrid) it is set to 10, since the model only needs to generate a short label.
- For cot strategies (cot and persona with cot hybrid) it is set to 400, since the model needs space to generate a reasoning chain before producing the final label.

pad_token_id: The pad_token_id is set to eos_token_id to ensure consistent behavior during batch generation.

enable_thinking: The Qwen3.5-4B model supports a thinking mode where it generates internal reasoning inside <think>...</think> blocks before producing the final response. This mode is explicitly disabled by setting enable_thinking=False for all experiments to keep the comparison fair across all three models. However it was observed that even with enable_thinking=False, the 4B model sometimes still generated <think>...</think> blocks, which are handled during output normalization as described in Section 5.5.

5.4. Model Card Sampling Parameters

For each model the recommended sampling parameters from the official Hugging Face model card are followed. The motivation behind this decision is that the producers of each model have already identified the optimal settings for their model, so following these parameters should give the best possible results. As discussed in Section 6.1, experiments were conducted both with and without these parameters to verify their impact on performance.

Figure 4 shows the sampling parameters used for each model and strategy combination:

Model	Strategy	do_sample	temperature	top_p	top_k
Qwen3.5-0.8B	all strategies	True	1.0	1.0	20
Qwen3.5-2B	all strategies	True	1.0	1.0	20
Qwen3.5-4B	zero-shot, few-shot, persona, persona + few-shot	True	0.7	0.8	20
Qwen3.5-4B	cot and Persona+cot	True	1.0	1.0	40

Figure 4: Sampling parameters per model and strategy from the official Hugging Face model cards

Note that the Qwen3.5-4B model uses two different sets of parameters: one for general tasks (zero-shot, few-shot, persona and persona with few-shot hybrid) and one for reasoning tasks (cot and persona with cot hybrid), as recommended by the model card.

5.5. Output Normalization

Since the models sometimes return the label with extra text around it, a normalization step is applied to extract the valid label from the raw model response. If no valid label is found the prediction is marked as INVALID. Two different normalization functions are used depending on the prompting strategy.

normalize_label() is used for all non-cot strategies. It applies the following steps in order:

1. Remove <think>...</think> blocks if present, since the 4B model sometimes generates these even with enable_thinking=False.
2. Take only the first line of the response and ignore everything after the first newline.
3. Check if the response is already a valid label (exact match).
4. If not, check if any valid label appears inside the response (substring match).
5. If no valid label is found return the INVALID marker.

normalize_label_cot() is used for cot and persona with cot hybrid strategies. cot responses contain reasoning text before the final label, so searching from the beginning of the response as normalize_label() does would produce incorrect results. For example, during reasoning the model might write "it's not a Clear Reply" and normalize_label() would incorrectly return Clear Reply because it found the label name first in the reasoning text. Instead, normalize_label_cot() searches from the end of the response for the "Label: X" pattern. So we have:

1. Remove <think>...</think> blocks if present.
2. Search from the end of the response for a line starting with "Label:" and extract the label from it.
3. If no "Label:" line is found fall back to normalize_label().

5.6. Invalid Prediction Handling

A prediction is marked as INVALID when the normalization function cannot find a valid label in the raw model response. For non-cot strategies this happens when normalize_label() cannot find a valid label. For cot strategies, normalize_label_cot() is used first and if it cannot find a valid label it calls normalize_label() as a fallback. Only when both functions fail to find a valid label the prediction is marked as INVALID. After each experiment run the number of invalid predictions is counted and reported as a percentage of the total predictions. Invalid predictions are kept in the predictions list and affect the evaluation metrics negatively since they never match any gold label.

5.7. Reproducibility

To ensure reproducibility a fixed seed of 42 is applied to all libraries that involve randomness.
A fixed pool of 1000 samples is created first from the training set using the fixed seed and stratification to preserve the class distribution (59.2% Ambivalent, 30.5% Clear Reply and 10.3% Clear Non-Reply). All experiments regardless of sample size are always drawn from within this same 1000 sample pool with stratification to preserve the class distribution. Thanks to the fixed seed, running the same experiment multiple times with the same sample size will always produce the exact same samples, ensuring that all comparisons across experiments are fair and reproducible.

6. Experiments

6.1. Experimental Setup

The goal of the experiments is to systematically compare all prompting strategies across all three model sizes, to identify the best performing combination and to investigate whether improving the existing prompts or adding new hybrid prompting strategies can further increase the overall performance. Experiments were conducted with six different sample sizes: 100, 300, 350, 400, 450 and 500 samples, all drawn from the fixed 1000 sample pool described in Section 5.7. Each experiment was run under two conditions: with and without the recommended sampling parameters from the official Hugging Face model cards, to assess their impact on performance.

oom Related Decisions: Three decisions were made during the experiments due to GPU memory constraints:

1. **500 samples not used:**
   At 500 samples the 4B model ran into oom errors. Fixing this would require reducing BATCH_SIZE from 4 to 2 or even 1, which would make inference impractically slow. Since the 4B model already required approximately 194 minutes to complete all strategies at 400 samples, running 500 samples with BATCH_SIZE 2 or 1 would take an impractical amount of time and was therefore not worth it.

2. **BATCH_SIZE reduced to 2 for Persona+Few-shot with 4B:**
   The persona with few-shot hybrid strategy produces longer inputs than other

strategies because it combines the persona system prompt with 3 labeled examples each containing a Question, Answer, Label and Reason field and all of these combined make the inputs longer. With BATCH_SIZE=4 this caused oom errors for the 4B model. The batch size was reduced to 2 only for this specific strategy and model combination so we can compare its results with the other models for this strategy.

3. **More few-shot examples not tested:**
   Adding more few-shot examples (2 or 3 per class instead of 1 per class) was avoided because the longer prompts would cause oom errors and would require reducing BATCH_SIZE from 4 to 2, roughly doubling the inference time. This is a limitation of the available GPU memory and not a fundamental limitation of the few-shot approach. Although adding more examples would likely improve performance, the trade-off between potentially better results and significantly longer inference time was not considered worth it, so this approach was not pursued.

6.2. Experiments Table

Rank	Exp	Samples	Model Card	Additional Changes	Best Model + Strategy	Val Weighted F1	Test Weighted F1
1	13	400	✅	Same improved prompts as Exp 11, max_new_tokens=400 for cot (all previous experiments used 150), added Persona+Few-shot and Persona+CoT hybrids	Qwen3.5-0.8B + few-shot	0.4247	—
1	11	400	✅	Improved few-shot (Reason field), improved persona prompt	Qwen3.5-0.8B + few-shot	0.4274	**0.4844**
2	3	400	✅	First run at 400 samples	Qwen3.5-2B + persona	0.4042	0.4634
3	6	300	✅	First run at 300 samples	Qwen3.5-0.8B + zero-shot	0.4082	0.4529
4	10	450	✅	First run at 450 samples	Qwen3.5-4B + zero-shot	0.3682	0.4405
5	12	350	✅	First run at 350 samples	Qwen3.5-2B + persona	0.4119	0.4257
6	1	100	❌	Baseline — no model card settings	Qwen3.5-0.8B + zero-shot	0.3027	0.4293
6	4	400	❌	Control for Exp 3 — no model card settings	Qwen3.5-0.8B + zero-shot	0.3141	0.4293
6	5	300	❌	Control for Exp 6 — no model card settings	Qwen3.5-0.8B + few-shot	0.3039	0.4293
6	9	450	❌	Control for Exp 10 — no model card settings	Qwen3.5-0.8B + few-shot	0.3200	0.4293
7	7	500	❌	4B OOM — only 0.8B and 2B completed	Qwen3.5-0.8B + few-shot	0.3296	—
7	8	500	✅	4B OOM — only 0.8B and 2B completed	Qwen3.5-2B + few-shot	0.3890	—
7	2	100	✅	Baseline — with model card settings	Qwen3.5-2B + few-shot	0.3966	0.2900

Figure 5: Summary of all 13 experiments ranked by Test Weighted F1 score

6.3. Individual Experiment Descriptions

6.3.1. Experiment 1 — Baseline without Model Card Settings (100 samples).

Experiment 1 is the baseline experiment. It uses 100 samples without the recommended sampling parameters from the official Hugging Face model cards and tests four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). Note that the persona prompt used in this experiment is the first version described in Section 4.4 (page 9).

The results show that zero-shot prompting achieved the best weighted F1 score of 0.3027 with the Qwen3.5-0.8B model. This is a surprising finding since one would expect larger models to outperform smaller ones. A possible explanation is that the smaller model was better able to follow the simple zero-shot instruction without being confused by additional context.

cot prompting performed very poorly across all three models. By inspecting the raw model outputs of the INVALID predictions in Experiment 10, it was observed that in all cases the model had generated a reasoning chain that exceeded the token limit of 150 tokens, causing the output to be cut off before reaching the required Label: <label> line at the end. The invalid rates were: 47.0% for Qwen3.5-0.8B, 72.0% for Qwen3.5-2B and 89.0% for Qwen3.5-4B. The larger the model the worse the invalid rate, suggesting that larger models tend to generate longer reasoning chains.

Persona prompting also performed poorly, achieving the lowest weighted F1 scores across all models. A possible explanation is that the role description focused on "evasion tactics" and "how politicians use language to avoid direct answers", which may have biased the model as confirmed by the per-class results in Table 2. As shown in the table, all three models have a very high Clear Non-Reply recall (0.30 for Qwen3.5-0.8B, 0.70 for Qwen3.5-2B and 0.90 for Qwen3.5-4B), confirming that the evasion-focused framing caused the models to over predict Clear Non-Reply. This bias in the persona prompt is discussed in detail in the error analysis subsection "How Different Prompting Methods Fail Differently" (Section 8.3).

Model	Class	Precision	Recall	F1
Qwen3.5-0.8B	Ambivalent	0.12	0.02	0.03
Qwen3.5-0.8B	Clear Non-Reply	0.04	0.30	0.08
Qwen3.5-0.8B	Clear Reply	0.32	0.26	0.29
Qwen3.5-2B	Ambivalent	0.44	0.12	0.19
Qwen3.5-2B	Clear Non-Reply	0.13	0.70	0.22
Qwen3.5-2B	Clear Reply	0.41	0.39	0.40
Qwen3.5-4B	Ambivalent	0.50	0.05	0.09
Qwen3.5-4B	Clear Non-Reply	0.11	0.90	0.20
Qwen3.5-4B	Clear Reply	0.50	0.23	0.31

Table 2: Per-class results for persona prompting in Experiment 1

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3027	0.2290
Qwen3.5-0.8B	few-shot	0.2647	0.2098
Qwen3.5-0.8B	cot	0.1277	0.1443
Qwen3.5-0.8B	persona	0.1140	0.1312
Qwen3.5-2B	zero-shot	0.2433	0.3330
Qwen3.5-2B	few-shot	0.1932	0.2737
Qwen3.5-2B	cot	0.1143	0.1261
Qwen3.5-2B	persona	0.2557	0.2674
Qwen3.5-4B	zero-shot	0.3024	0.3361
Qwen3.5-4B	few-shot	0.2251	0.3049
Qwen3.5-4B	cot	0.0944	0.1160
Qwen3.5-4B	persona	0.1709	0.2011

Table 3: Experiment 1 results — all model and strategy combinations

Due to the small sample size of 100, the results of this experiment may not be fully reliable or representative of the true model performance. For this reason subsequent experiments were conducted with larger sample sizes to obtain more statistically reliable results and to be more confident that the observed differences between models and strategies are truly genuine.

6.3.2. Experiment 2 — Baseline with Model Card Settings (100 samples).

Experiment 2 is identical to Experiment 1 but with the recommended sampling parameters from the official Hugging Face model cards applied. It uses the same 100 samples and tests the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

The best combination in this experiment was Qwen3.5-2B + few-shot with a weighted F1 of 0.3966. A possible explanation is that with the model card sampling parameters

applied, the 2B model was better able to leverage the labeled examples provided in the few-shot prompt to understand the classification task. The model card parameters help the 2B model generate more consistent outputs and the labeled examples give it concrete guidance on how to classify each label, making this combination particularly effective.

Applying the model card settings improved the results for most strategies. The best weighted F1 score shifted from Qwen3.5-0.8B + zero-shot (0.3027) in Experiment 1 to Qwen3.5-2B + few-shot (0.3966) in this experiment, an improvement of 0.0939. This suggests that the model card sampling parameters help larger models perform better, particularly with few-shot prompting.

- **Zero-shot:** Improved for all models. The most notable improvement was for Qwen3.5-4B which went from 0.3024 to 0.3865, a gain of 0.0841. Qwen3.5-0.8B improved from 0.3027 to 0.3451 and Qwen3.5-2B improved from 0.2433 to 0.2984. This suggests that the model card sampling parameters help all models generate more consistent and accurate labels even in the simplest prompting setting.
- **Few-shot:** Improved significantly for Qwen3.5-2B (from 0.1932 to 0.3966, a gain of 0.2034) and for Qwen3.5-0.8B (from 0.2647 to 0.3439). However Qwen3.5-4B dropped slightly from 0.2251 to 0.2961. The large improvement for Qwen3.5-2B suggests that this model benefits the most from the model card parameters when given labeled examples to follow.
- **cot:** Improved slightly for all models but remained very poor due to the token limit issue. The invalid rates of 55.0% for Qwen3.5-0.8B, 64.0% for Qwen3.5-2B and 91.0% for Qwen3.5-4B, confirming that the token limit of 150 is the main bottleneck for cot regardless of the sampling parameters used.
- **Persona:** Improved for all models. The most notable improvement was for Qwen3.5-0.8B which went from 0.1140 to 0.3476, a gain of 0.2336. Qwen3.5-2B improved from 0.2557 to 0.3530 and Qwen3.5-4B improved from 0.1709 to 0.2302. This suggests that the model card parameters help the models better follow the explicit methodology defined in the persona prompt.

However the test set performance dropped significantly from 0.4293 (Experiment 1) to 0.2900 (Experiment 2), despite the higher validation F1. This large gap between validation and test performance suggests that with only 100 samples the results are not reliable and the model overfits to the small validation set.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3451	0.2053
Qwen3.5-0.8B	few-shot	0.3439	0.1963
Qwen3.5-0.8B	cot	0.1520	0.1526
Qwen3.5-0.8B	persona	0.3476	0.2104
Qwen3.5-2B	zero-shot	0.2984	0.2788
Qwen3.5-2B	few-shot	0.3966	0.2966
Qwen3.5-2B	cot	0.1765	0.1768
Qwen3.5-2B	persona	0.3530	0.2461
Qwen3.5-4B	zero-shot	0.3865	0.3945
Qwen3.5-4B	few-shot	0.2961	0.3489
Qwen3.5-4B	cot	0.0837	0.0957
Qwen3.5-4B	persona	0.2302	0.2214

Table 4: Experiment 2 results — all model and strategy combinations

6.3.3. Experiment 3 — 400 Samples with Model Card Settings.

Experiment 3 is the first experiment conducted with 400 samples and with the recommended sampling parameters from the official Hugging Face model cards. It tests the same four prompting strategies as Experiments 1 and 2: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

Moving from 100 to 400 samples produced more stable and reliable results. The best weighted F1 score in this experiment was achieved by Qwen3.5-2B + persona (0.4042), which is higher than the best result from Experiment 2 (0.3966). This improvement is consistent with our expectation that larger sample sizes lead to more reliable results.

The best combination in this experiment was Qwen3.5-2B + persona with a weighted F1 of 0.4042. A possible explanation is that the persona prompt provides the 2B model with an explicit three-step classification methodology, which guides the model to systematically analyze each question-answer pair before classifying it. The 2B model appears to benefit more from this structured methodology compared to the smaller 0.8B model which may lack the capacity to fully follow the explicit steps, and compared to the larger 4B model which despite its larger size does not always outperform smaller models on this task, suggesting that model size alone is not the determining factor for performance.

- **Zero-shot:** Results became more stable across all models. Qwen3.5-4B achieved the highest zero-shot score of 0.3575, followed by Qwen3.5-0.8B (0.3690) and Qwen3.5-2B (0.3118).
- **Few-shot:** Qwen3.5-0.8B improved significantly to 0.3955, while Qwen3.5-2B achieved 0.3747 and Qwen3.5-4B 0.3265. This improvement compared to Experiment 2 is consistent with the expectation that larger sample sizes produce more reliable results, as we can get more reliable results on how each model truly performs with each prompting strategy.

- **cot:** Remained very poor due to the token limit issue. The invalid rates were 54.8% for Qwen3.5-0.8B, 61.5% for Qwen3.5-2B and 91.5% for Qwen3.5-4B, consistent with the pattern observed in Experiments 1 and 2.
- **Persona:** Qwen3.5-2B + persona achieved the best result of this experiment with a weighted F1 of 0.4042, an improvement over the best persona result from Experiment 2 which was Qwen3.5-2B + persona (0.3530). This improvement is consistent with our expectation that larger sample sizes lead to more reliable results. However it is worth noting that Qwen3.5-4B + persona shows an extreme bias towards predicting Clear Non-Reply (recall=0.80), as shown in Table 5, confirming the pattern observed in Experiment 1 (Table 2) that the persona prompt causes the models to over predict Clear Non-Reply. This bias is discussed in detail in the error analysis subsection "How Different Prompting Methods Fail Differently" (Section 8.3).

Class	Precision	Recall	F1
Ambivalent	0.45	0.14	0.21
Clear Non-Reply	0.13	0.80	0.22
Clear Reply	0.56	0.30	0.39

Table 5: Per-class results for Qwen3.5-4B + persona Experiment 3

The test set weighted F1 of 0.4634 for Qwen3.5-2B + persona is significantly higher than Experiment 2 (0.2900), confirming that 400 samples produces more reliable and generalizable results than 100 samples.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3690	0.2100
Qwen3.5-0.8B	few-shot	0.3955	0.2218
Qwen3.5-0.8B	cot	0.1739	0.1579
Qwen3.5-0.8B	persona	0.3363	0.2018
Qwen3.5-2B	zero-shot	0.3118	0.2464
Qwen3.5-2B	few-shot	0.3747	0.2592
Qwen3.5-2B	cot	0.2533	0.1978
Qwen3.5-2B	persona	0.4042	0.2517
Qwen3.5-4B	zero-shot	0.3575	0.3576
Qwen3.5-4B	few-shot	0.3265	0.3468
Qwen3.5-4B	cot	0.0769	0.0795
Qwen3.5-4B	persona	0.2654	0.2728

Table 6: Experiment 3 results — all model and strategy combinations

6.3.4. Experiment 4 — 400 Samples without Model Card Settings.

Experiment 4 uses the same 400 samples as Experiment 3 but without the recommended sampling parameters from the official Hugging Face model cards. As shown

in Experiments 1 and 2, the model card settings consistently improved performance across all models and strategies on the validation set, but the test weighted F1 dropped significantly from 0.4293 (Experiment 1) to 0.2900 (Experiment 2), suggesting that with only 100 samples the results are not reliable and the model overfits to the small validation set. Experiment 4 investigates whether the same pattern appears at 400 samples. The best combination in this experiment was Qwen3.5-0.8B + zero-shot with a weighted F1 of 0.3141. This is consistent with Experiment 1 where the same combination was also the best without model card settings, suggesting that without the recommended sampling parameters the smallest model with the simplest prompting strategy is the most robust combination, producing the most stable and best results across both experiments which do not follow the model card parameters.

Comparing Experiment 4 (without model card settings) to Experiment 3 (with model card settings) at the same 400 samples, the model card settings consistently produced better results across all models and strategies:

- **Zero-shot:** Without model card settings all models performed worse. Qwen3.5-0.8B dropped from 0.3690 to 0.3141, Qwen3.5-2B dropped from 0.3118 to 0.2194 and Qwen3.5-4B dropped from 0.3575 to 0.2947. This confirms that the model card parameters help all models generate more consistent and accurate labels even in the simplest prompting setting.
- **Few-shot:** Without model card settings all models performed worse. Qwen3.5-0.8B dropped from 0.3955 to 0.3140, Qwen3.5-2B dropped from 0.3747 to 0.2172 and Qwen3.5-4B dropped from 0.3265 to 0.2278. This suggests that the model card parameters are particularly important for few-shot prompting.
- **cot:** Remained very poor in both cases due to the token limit issue. The invalid rates were 51.2% for Qwen3.5-0.8B, 65.5% for Qwen3.5-2B and 85.5% for Qwen3.5-4B, confirming that the token limit of 150 is the main problem for cot regardless of the sampling parameters used.
- **Persona:** Without model card settings all models performed significantly worse. Qwen3.5-0.8B dropped from 0.3363 to 0.1486, Qwen3.5-2B dropped from 0.4042 to 0.2809 and Qwen3.5-4B dropped from 0.2654 to 0.1783. The large drop for persona suggests that without the correct sampling parameters the models struggle to follow the explicit methodology defined in the persona prompt.

Unlike Experiments 1 and 2 where the experiment without model card settings (Experiment 1) achieved a higher test weighted F1 of 0.4293 compared to 0.2900 with model card settings (Experiment 2), at 400 samples the pattern is reversed. Experiment 3 with model card settings achieved a higher test weighted F1 of 0.4634 compared to 0.4293 without model card settings in this experiment, confirming that at larger sample sizes the model card settings consistently improve performance on both the validation and test sets.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3141	0.2268
Qwen3.5-0.8B	few-shot	0.3140	0.2523
Qwen3.5-0.8B	cot	0.1228	0.1225
Qwen3.5-0.8B	persona	0.1486	0.1627
Qwen3.5-2B	zero-shot	0.2194	0.2946
Qwen3.5-2B	few-shot	0.2172	0.2753
Qwen3.5-2B	cot	0.1405	0.1422
Qwen3.5-2B	persona	0.2809	0.2825
Qwen3.5-4B	zero-shot	0.2947	0.3209
Qwen3.5-4B	few-shot	0.2278	0.2895
Qwen3.5-4B	cot	0.1271	0.1463
Qwen3.5-4B	persona	0.1783	0.2140

Table 7: Experiment 4 results — all model and strategy combinations

6.3.5. Experiment 5 — 300 Samples without Model Card Settings.

Experiment 5 uses 300 samples without the recommended sampling parameters from the official Hugging Face model cards. It tests the same four prompting strategies as the previous experiments: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

The best combination in this experiment was Qwen3.5-0.8B + few-shot with a weighted F1 of 0.3039. This is consistent with the pattern observed in Experiments 1 and 4 where without model card settings the smaller models with simpler strategies tend to be the most robust combination, producing the most stable and best results across experiments which do not follow the model card parameters.

The results are consistent with the pattern observed in Experiment 4. Without model card settings all models and strategies performed worse compared to their counterparts with model card settings. cot remained very poor with invalid rates of 48.3% for Qwen3.5-0.8B, 69.7% for Qwen3.5-2B and 86.0% for Qwen3.5-4B, again confirming that the token limit of 150 is the main problem for cot regardless of the sampling parameters used.

The test set weighted F1 of 0.4293 for Qwen3.5-0.8B + zero-shot is identical to Experiments 1, 4 and 9, confirming the consistent pattern that without model card settings the best combination on the test set is always Qwen3.5-0.8B + zero-shot.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.2764	0.1991
Qwen3.5-0.8B	few-shot	0.3039	0.2509
Qwen3.5-0.8B	cot	0.1182	0.1243
Qwen3.5-0.8B	persona	0.1279	0.1510
Qwen3.5-2B	zero-shot	0.2236	0.3023
Qwen3.5-2B	few-shot	0.2265	0.2980
Qwen3.5-2B	cot	0.1439	0.1392
Qwen3.5-2B	persona	0.2627	0.2748
Qwen3.5-4B	zero-shot	0.2966	0.3274
Qwen3.5-4B	few-shot	0.2278	0.2879
Qwen3.5-4B	cot	0.1274	0.1323
Qwen3.5-4B	persona	0.1565	0.2024

Table 8: Experiment 5 results — all model and strategy combinations

6.3.6. Experiment 6 — 300 Samples with Model Card Settings.

Experiment 6 uses 300 samples with the recommended sampling parameters from the official Hugging Face model cards. It is the counterpart of Experiment 5 which used the same 300 samples without model card settings. It tests the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).
The best combination in this experiment was Qwen3.5-0.8B + zero-shot with a weighted F1 of 0.4082, closely followed by Qwen3.5-2B + persona with 0.4055 i.e. a difference of only 0.0027 which is negligible. This suggests that at 300 samples with model card settings both zero-shot with the smallest model and persona with the medium model perform essentially equally well. A possible explanation is that the smaller model was better able to follow the zero-shot instruction without being confused by additional context, while the persona prompt provides the 2B model with an explicit three-step classification methodology which guides it to systematically analyze each question-answer pair before classifying. The 2B model appears to benefit more from this structured methodology compared to the smaller 0.8B model which may lack the capacity to fully follow the explicit steps. Both approaches reach a similar level of performance, suggesting that at 300 samples a simple but well-tuned strategy can match a more complex but structured one.

Comparing Experiment 6 (with model card settings) to Experiment 5 (without model card settings) at the same 300 samples, the model card settings consistently improved results across all models and strategies. The most notable improvements were for Qwen3.5-0.8B + zero-shot (from 0.2764 to 0.4082, a gain of 0.1318) and Qwen3.5-2B + persona (from 0.2627 to 0.4055, a gain of 0.1428). cot remained very poor with invalid rates of 51.7% for Qwen3.5-0.8B, 64.0% for Qwen3.5-2B and 91.7% for Qwen3.5-4B.

The test set weighted F1 of 0.4529 for Qwen3.5-0.8B + zero-shot is higher than Experiment 5 (0.4293), confirming again that the model card settings improve performance

on both the validation and test sets at 300 samples.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.4082	0.2312
Qwen3.5-0.8B	few-shot	0.3890	0.2194
Qwen3.5-0.8B	cot	0.1618	0.1223
Qwen3.5-0.8B	persona	0.3500	0.2228
Qwen3.5-2B	zero-shot	0.3411	0.2534
Qwen3.5-2B	few-shot	0.3026	0.2280
Qwen3.5-2B	cot	0.1516	0.1281
Qwen3.5-2B	persona	0.4055	0.2656
Qwen3.5-4B	zero-shot	0.3261	0.3400
Qwen3.5-4B	few-shot	0.3071	0.3305
Qwen3.5-4B	cot	0.0837	0.1046
Qwen3.5-4B	persona	0.2171	0.2297

Table 9: Experiment 6 results — all model and strategy combinations

6.3.7. Experiments 7 and 8 — 500 Samples (without and with model card settings).

Experiments 7 and 8 attempt to scale up to 500 samples, with Experiment 7 running without model card settings and Experiment 8 running with model card settings. Both experiments test the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

However the Qwen3.5-4B model ran into oom errors at 500 samples and could not complete the inference. The root cause is that the 4B model requires approximately 12GB of the available 14.56GB GPU memory just to load, leaving only about 2.5GB for inference. Few-shot prompts are longer than zero-shot because the zero-shot has just the question and answer, but few-shot has 3 labeled examples plus the question and answer, so with only 1.91GB of free memory there was not enough GPU memory to process the batches. Fixing this would require reducing BATCH_SIZE from 4 to 2, which would roughly double the inference time. Since the 4B model already required approximately 194 minutes to complete all strategies at 400 samples, this was not considered worth it. Therefore 500 samples was not used for the 4B model and these two experiments were completed only for Qwen3.5-0.8B and Qwen3.5-2B. This represents the practical memory ceiling under the available GPU constraints.

6.3.8. Experiment 9 — 450 Samples without Model Card Settings.

Experiment 9 uses 450 samples without the recommended sampling parameters from the official Hugging Face model cards. It tests the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

As the experiments progressed from 100 to 300 to 400 samples we consistently obtained better weighted validation F1 scores with each increase in sample size. Since 500 samples was found to be the practical memory ceiling as described in Experiments 7 and 8, 450 samples was chosen as the next step to investigate whether we can get even better results.

The results follow the same pattern observed in Experiments 4 and 5. Without model card settings the best combination was Qwen3.5-0.8B + few-shot with a weighted F1 of 0.3200, consistent with the pattern that without the recommended sampling parameters the smallest model with a simple strategy is the most robust combination. The test weighted F1 of 0.4293 is identical to Experiments 1, 4 and 5, confirming the consistent pattern that without model card settings the best combination on the test set is always Qwen3.5-0.8B + zero-shot with a test weighted F1 of 0.4293 regardless of the sample size.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3205	0.2270
Qwen3.5-0.8B	few-shot	0.3200	0.2540
Qwen3.5-0.8B	cot	0.1297	0.1305
Qwen3.5-0.8B	persona	0.1391	0.1563
Qwen3.5-2B	zero-shot	0.2211	0.2857
Qwen3.5-2B	few-shot	0.2229	0.2792
Qwen3.5-2B	cot	0.1505	0.1523
Qwen3.5-2B	persona	0.2921	0.2901
Qwen3.5-4B	zero-shot	0.2964	0.3202
Qwen3.5-4B	few-shot	0.2393	0.2988
Qwen3.5-4B	cot	0.1158	0.1239
Qwen3.5-4B	persona	0.1876	0.2221

Table 10: Experiment 9 results — all model and strategy combinations

6.3.9. Experiment 10 — 450 Samples with Model Card Settings.

Experiment 10 uses 450 samples with the recommended sampling parameters from the official Hugging Face model cards. It is the counterpart of Experiment 9 which used the same 450 samples without model card settings. It tests the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

As the experiments progressed from 100 to 300 to 400 samples we consistently obtained better weighted validation F1 scores with each increase in sample size. Since 500 samples was found to be the practical memory ceiling as described in Experiments 7 and 8, 450 samples was chosen as the next step to investigate whether we can get even better results.

The best combination in this experiment was Qwen3.5-4B + zero-shot with a weighted F1 of 0.3682. This is surprising since in all previous experiments with model card

settings the best combination was never the 4B model with zero-shot.
Comparing Experiment 10 to Experiment 3 (400 samples with model card settings), the best weighted validation F1 dropped from 0.4042 to 0.3682, a decrease of 0.0360. This confirms that 400 samples appears to be the sweet spot for this task, as increasing the sample size further to 450 samples did not improve and actually slightly hurt performance.

The test set weighted F1 of 0.4405 for Qwen3.5-0.8B + few-shot is lower than Experiment 3 (0.4634), further confirming that 400 samples produces better generalizable results than 450 samples for this task.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3847	0.2175
Qwen3.5-0.8B	few-shot	0.3891	0.2244
Qwen3.5-0.8B	cot	0.1991	0.1480
Qwen3.5-0.8B	persona	0.3470	0.2044
Qwen3.5-2B	zero-shot	0.3488	0.2694
Qwen3.5-2B	few-shot	0.3802	0.2693
Qwen3.5-2B	cot	0.1992	0.1631
Qwen3.5-2B	persona	0.3703	0.2280
Qwen3.5-4B	zero-shot	0.3682	0.3683
Qwen3.5-4B	few-shot	0.2781	0.3069
Qwen3.5-4B	cot	0.1107	0.1301
Qwen3.5-4B	persona	0.2260	0.2374

Table 11: Experiment 10 results — all model and strategy combinations

6.3.10. Experiment 11 (Best Experiment) — Improved Few-shot and Persona Prompts (400 samples, with model card settings).

Since we found the sweet spot for the best number of samples in Experiment 3 (400 samples with model card settings), Experiment 11 now investigates whether improving the existing prompts can further increase the overall performance. It uses the same 400 samples and model card settings as Experiment 3 and tests the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). Two targeted prompt improvements were made compared to all previous experiments:

- **Few-shot:** A Reason field was added to each labeled example explaining why that label was chosen, changing the pattern from Question → Answer → Label to Question → Answer → Label → Reason. The Reason field was added because simply showing Question → Answer → Label may not give the model enough context to understand the classification logic. By explicitly explaining why each label was chosen, the model gets a clearer picture of what distinguishes each class, which helps it make more accurate classifications on unseen examples. This improvement is described also in Section 4.2.

- **Persona:** the persona prompt was improved in two ways: the role description was changed from "analyzing evasion tactics in political interviews" to "analyzing political interview responses", and Step 2 of the methodology was made more explicit by adding "directly and fully" to the description. The role description was changed because the original framing focused on evasion tactics which may have biased the model. The more neutral description of "analyzing political interview responses" avoids this bias by not priming the model to look for evasion. Step 2 was made more explicit by adding "directly and fully" to make it clearer to the model that a partial or vague answer should not be classified as a Clear Reply, which helps reduce confusion between the Ambivalent and Clear Reply classes. These improvements are described also in Section 4.4 (page 10).

The best combination in this experiment was Qwen3.5-0.8B + few-shot with a weighted F1 of 0.4274, an improvement of 0.0232 over the best result from Experiment 3 (Qwen3.5-2B + persona, 0.4042). This is the highest validation weighted F1 score achieved across all experiments. A possible explanation is that adding the Reason field to each few-shot example gives the 0.8B model concrete reasoning behind each label, helping it better understand the classification logic. The 0.8B model appears to benefit more from this explicit reasoning guidance compared to the larger models.

Comparing Experiment 11 to Experiment 3 at the same 400 samples with model card settings:

- **Zero-shot:** Results remained essentially the same since the zero-shot prompt was not changed.
- **Few-shot:** Improved significantly for Qwen3.5-0.8B (from 0.3955 to 0.4274, a gain of 0.0319) and for Qwen3.5-2B (from 0.3747 to 0.4037, a gain of 0.0290). However Qwen3.5-4B dropped from 0.3265 to 0.2876. This confirms that adding the Reason field to the few-shot examples significantly helps the smaller models understand the classification logic. By explicitly explaining why each label was chosen, the model gets a clearer picture of what distinguishes each class. The 4B model however does not benefit from it in the same way.
- **cot:** Remained very poor due to the token limit of 150 tokens. The invalid rates were 58.5% for Qwen3.5-0.8B, 92.5% for Qwen3.5-2B and 87.2% for Qwen3.5-4B. Notably the invalid rate for Qwen3.5-2B increased significantly compared to Experiment 3 (from 61.5% to 92.5%).
- **Persona:** The improved persona prompt helped Qwen3.5-4B (from 0.2654 to 0.3438) but slightly hurt Qwen3.5-0.8B (from 0.3363 to 0.2958) and Qwen3.5-2B (from 0.4042 to 0.3916). The role description was changed because the original framing focused on evasion tactics which may have biased the model. The more neutral description of "analyzing political interview responses" avoids this bias by not priming the model to look for evasion. Step 2 was made more explicit by adding "directly and fully" to make it clearer to the model that a partial or vague answer should not be classified as a Clear Reply, which helps reduce confusion between the Ambivalent and Clear Reply classes. These changes appear to benefit the larger 4B model more, while the smaller models may find the updated methodology harder to follow consistently.

The test set weighted F1 of 0.4844 for Qwen3.5-0.8B + few-shot is the highest test score achieved across all experiments, confirming that the prompt improvements in Experiment 11 produced the best generalizable results. This represents a significant improvement over the previous best test score of 0.4634 from Experiment 3, a gain of 0.0210.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3690	0.2100
Qwen3.5-0.8B	few-shot	0.4274	0.2429
Qwen3.5-0.8B	cot	0.1657	0.1445
Qwen3.5-0.8B	persona	0.2958	0.1913
Qwen3.5-2B	zero-shot	0.3445	0.2560
Qwen3.5-2B	few-shot	0.4037	0.2887
Qwen3.5-2B	cot	0.0679	0.0737
Qwen3.5-2B	persona	0.3916	0.2586
Qwen3.5-4B	zero-shot	0.3548	0.3490
Qwen3.5-4B	few-shot	0.2876	0.3167
Qwen3.5-4B	cot	0.1176	0.1294
Qwen3.5-4B	persona	0.3438	0.3404

Table 12: Experiment 11 results — all model and strategy combinations

6.3.11. Experiment 12 — 350 Samples with Model Card Settings.

Experiment 12 uses 350 samples with the recommended sampling parameters from the official Hugging Face model cards. It tests the same four prompting strategies: system_zero_shot (Section 4.1), build_few_shot (Section 4.2), system_cot (Section 4.3) and system_role (Section 4.4). The persona prompt used is still the first version described in Section 4.4 (page 9).

The best combination in this experiment was Qwen3.5-2B + persona with a weighted F1 of 0.4119, consistent with the pattern observed in Experiment 3 where the same combination was also the best. The test set weighted F1 of 0.4257 for Qwen3.5-2B + persona is lower than Experiment 3 (0.4634), confirming again that 400 samples produces better generalizable results than 350 samples.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.4065	0.2345
Qwen3.5-0.8B	few-shot	0.3493	0.2008
Qwen3.5-0.8B	cot	0.1833	0.1651
Qwen3.5-0.8B	persona	0.3856	0.2306
Qwen3.5-2B	zero-shot	0.3442	0.2499
Qwen3.5-2B	few-shot	0.3847	0.2732
Qwen3.5-2B	cot	0.2052	0.1600
Qwen3.5-2B	persona	0.4119	0.2634
Qwen3.5-4B	zero-shot	0.3098	0.3098
Qwen3.5-4B	few-shot	0.2814	0.3144
Qwen3.5-4B	cot	0.0720	0.0796
Qwen3.5-4B	persona	0.2518	0.2557

Table 13: Experiment 12 results — all model and strategy combinations

6.3.12. Experiment 13 — Additional Hybrid Prompting Strategies and Increased CoT Token Limit (400 samples, with model card settings).

Experiment 13 is the final experiment and builds directly on Experiment 11. It uses the same 400 samples, model card settings and the same two prompt improvements from Experiment 11 (improved few-shot with Reason field and improved persona prompt). Two additional changes were made compared to Experiment 11:

- **Hybrid strategies added:** Two new hybrid prompting strategies were added: persona with few-shot hybrid (Section 4.5) and persona with CoT hybrid (Section 4.6). The idea behind both hybrids is to take the best elements from each strategy and combine them, hoping to achieve better results than either strategy alone. The persona with few-shot hybrid gives the model both the expert context and methodology from the persona prompt and the concrete labeled examples with reasoning from the few-shot prompt. The persona with CoT hybrid gives the model both the expert context and methodology from the persona prompt and the step-by-step reasoning from CoT prompting. The goal was to investigate whether these combinations could beat the best result from Experiment 11 (Qwen3.5-0.8B + few-shot, 0.4274).
- **CoT token limit increased:** The max_new_tokens for CoT strategies (CoT and persona with CoT hybrid) was increased from 150 to 400. As observed in Experiment 11 the CoT strategy produced very high INVALID prediction rates (58.5% for Qwen3.5-0.8B, 92.5% for Qwen3.5-2B and 87.2% for Qwen3.5-4B) because the model generated reasoning chains that exceeded the 150 token limit, causing the output to be cut off before reaching the required Label: <label> line. By inspecting the raw model outputs of the INVALID predictions it was confirmed that in all cases the text was cut off mid reasoning without producing the final label. Increasing max_new_tokens to 400 significantly reduced the INVALID rates as shown in Section 4.3, and more correct labels were produced. However even with more valid predictions the CoT strategy did not outperform the best strategy

from Experiment 11. Increasing max_new_tokens further to 600 was not considered worth it for two reasons: first, the improvement in INVALID rates from 150 to 400 was already very large and we expect diminishing returns going from 400 to 600 and second, the additional inference time would be significant since CoT strategies already dominate the total inference time, as discussed in Section 4.3.

The best combination in this experiment was Qwen3.5-0.8B + few-shot with a weighted F1 of 0.4247, which is essentially identical to Experiment 11 (0.4274) with a difference of only 0.0027. This difference is within the expected range of GPU non-determinism caused by the different max_new_tokens settings between the two experiments (150 in Experiment 11 vs 400 in Experiment 13), which changes the GPU memory state between strategies. This confirms that the best strategy remains Qwen3.5-0.8B + few-shot regardless of the additional changes made in this experiment.

Comparing Experiment 13 to Experiment 11 at the same 400 samples with model card settings:

- **Zero-shot:** Results remained essentially the same since the zero-shot prompt was not changed.
- **Few-shot:** Remained the best strategy overall with Qwen3.5-0.8B achieving 0.4247, closely matching Experiment 11 (0.4274). Qwen3.5-2B achieved 0.3697 and Qwen3.5-4B achieved 0.3161. The small differences are consistent with GPU non-determinism rather than any meaningful change in performance.
- **CoT:** The most significant change in this experiment. Increasing max_new_tokens from 150 to 400 dramatically reduced the INVALID prediction rates: from 58.5% to 18.0% for Qwen3.5-0.8B, from 92.5% to 22.2% for Qwen3.5-2B and from 87.2% to 23.0% for Qwen3.5-4B, as discussed in detail in Section 4.3. Despite this significant reduction in INVALID predictions, the CoT weighted F1 scores did not improve enough to beat the best strategy. Qwen3.5-0.8B CoT achieved 0.2735, Qwen3.5-2B achieved 0.2433 and Qwen3.5-4B achieved 0.3534. This suggests that even when the models can complete their reasoning chains, the step-by-step reasoning does not lead to better classifications than the simpler few-shot strategy with the Reason field. Increasing max_new_tokens further to 600 was not pursued as discussed in Section 4.3 (the time cost would be too high and the improvement in INVALID rates would likely be much smaller than the jump from 150 to 400).
- **Persona:** Results were consistent with Experiment 11. Qwen3.5-0.8B achieved 0.2806, Qwen3.5-2B achieved 0.3827 and Qwen3.5-4B achieved 0.3349.
- **Persona+Few-shot:** The first hybrid strategy combined the improved persona system prompt with the few-shot labeled examples with Reason field. Qwen3.5-0.8B achieved 0.3934, Qwen3.5-2B achieved 0.3477 and Qwen3.5-4B achieved 0.3375. Despite combining the best elements of both strategies, the hybrid did not outperform the simple few-shot strategy (0.4247). A possible explanation is that the persona system prompt adds complexity that interferes with the model's ability to leverage the labeled examples effectively, while the simple zero-shot system prompt used in regular few-shot provides a cleaner context for the examples to be followed.

- **Persona+CoT:** The second hybrid strategy combined the improved persona system prompt with step-by-step CoT reasoning. Qwen3.5-0.8B achieved 0.3120, Qwen3.5-2B achieved 0.4003 and Qwen3.5-4B achieved 0.3129. Again the hybrid did not outperform the best strategy. The INVALID rates were significantly lower than in Experiment 11 CoT due to the increased token limit: 11.2% for Qwen3.5-0.8B, 10.2% for Qwen3.5-2B and 39.0% for Qwen3.5-4B.

The test set result for Experiment 13 was not saved. However since the validation weighted F1 of 0.4247 is essentially identical to Experiment 11 (0.4274), with a difference of only 0.0027 which is within the expected range of GPU non-determinism caused by the different max_new_tokens settings between the two experiments (150 in Experiment 11 vs 400 in Experiment 13), we expect the test weighted F1 to be approximately 0.4844.

Model	Strategy	Weighted F1	Macro F1
Qwen3.5-0.8B	zero-shot	0.3951	0.2148
Qwen3.5-0.8B	few-shot	0.4247	0.2494
Qwen3.5-0.8B	cot	0.2735	0.1966
Qwen3.5-0.8B	persona	0.2806	0.1723
Qwen3.5-0.8B	persona+few-shot	0.3934	0.2134
Qwen3.5-0.8B	persona+CoT	0.3120	0.2226
Qwen3.5-2B	zero-shot	0.3491	0.2641
Qwen3.5-2B	few-shot	0.3697	0.2548
Qwen3.5-2B	cot	0.2433	0.2059
Qwen3.5-2B	persona	0.3827	0.2603
Qwen3.5-2B	persona+few-shot	0.3477	0.2386
Qwen3.5-2B	persona+CoT	0.4003	0.2610
Qwen3.5-4B	zero-shot	0.3296	0.3406
Qwen3.5-4B	few-shot	0.3161	0.3461
Qwen3.5-4B	cot	0.3534	0.2817
Qwen3.5-4B	persona	0.3349	0.3245
Qwen3.5-4B	persona+few-shot	0.3375	0.3350
Qwen3.5-4B	persona+CoT	0.3129	0.2611

Table 14: Experiment 13 results — all model and strategy combinations

7. Best Experiment Test Weighted F1 Score and Validation F1 Score

The best performing system across all experiments was Qwen3.5-0.8B + few-shot prompting from Experiment 11 (Section 6.3.10), achieving the highest test weighted F1 score of 0.4844 and the highest validation weighted F1 score of 0.4274 across all 13 experiments. This combination uses the improved few-shot prompt with the Reason field added to each labeled example and the model card sampling parameters, trained on 400 samples. Experiment 13 (Section 6.3.12) confirmed this result with an essentially identical validation weighted F1 of 0.4247, a difference of only 0.0027 which is within the expected range of GPU non-determinism. A detailed discussion of why this combination performed best is provided in Sections 6.3.10 and 6.3.12.

8. Error Analysis

8.1. Which Class is Hardest to Predict

To analyze which class is hardest to predict we use the per-class results from the best performing system across all 13 experiments, which is Qwen3.5-0.8B + few-shot from Experiment 11 (Section 6.3.10), achieving a test weighted F1 of 0.4844. The per-class breakdown on the test set is shown in Table 15:

Set	Class	Precision	Recall	F1
Validation	Ambivalent	0.59	0.51	0.55
Validation	Clear Non-Reply	0.10	0.20	0.13
Validation	Clear Reply	0.32	0.27	0.29
Test	Ambivalent	0.67	0.51	0.58
Test	Clear Non-Reply	0.11	0.35	0.17
Test	Clear Reply	0.34	0.32	0.33

Table 15: Per-class results of the best system (Qwen3.5-0.8B + few-shot, Experiment 11) on the validation and test sets

Clear Non-Reply is by far the hardest class to predict, achieving an F1 score of only 0.13 on the validation set and 0.17 on the test set. The very low precision (0.10 on validation, 0.11 on test) indicates that the model predicts Clear Non-Reply too broadly i.e. when it does predict this class it is wrong approximately 89% of the time on both sets. There are two main reasons for this:

- **Low support:** Clear Non-Reply is severely underrepresented in both the validation set (41 samples, 10.3%) and the test set (23 samples, 7.47%). This means that even a small number of misclassifications has a large impact on the F1 score for this class.
- **Confusion with Ambivalent:** As shown in Figure 6, the majority of Clear Non-Reply misclassifications (25 out of 30, i.e. 83%) are predicted as Ambivalent, suggesting the model tends to default to the dominant class when it is uncertain.

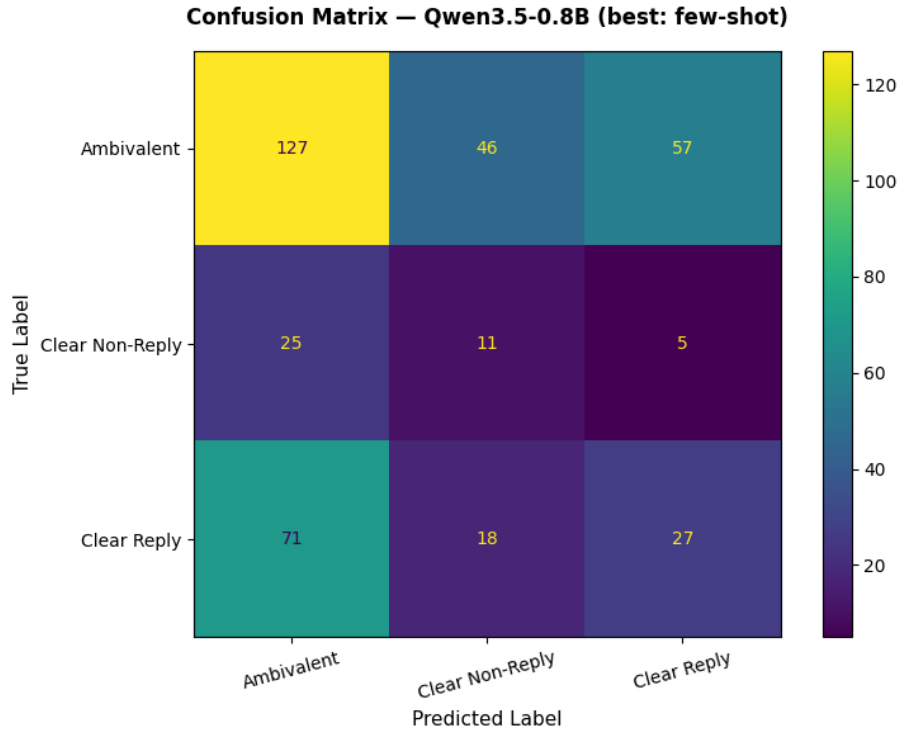


Figure 6: Confusion matrix for the best system (Qwen3.5-0.8B + few-shot, Experiment 11) on the validation set

Ambivalent is the easiest class to predict with an F1 of 0.55 on the validation set and 0.58 on the test set. This is expected since it is the dominant class in both the validation set (59.2%) and the test set (66.9%), giving the model the most exposure to this class. However the precision of 0.59 and recall of 0.51 on the validation set suggest that the model still misses a significant number of Ambivalent instances. As shown in Figure 6, out of the 110 misclassified Ambivalent instances, 57 are predicted as Clear Reply and 46 as Clear Non-Reply.

Clear Reply sits in between with an F1 of 0.29 on the validation set and 0.33 on the test set. The balanced precision and recall suggest that the model has a consistent but limited ability to identify this class. As shown in Figure 6, out of the 89 misclassified Clear Reply instances, 71 are predicted as Ambivalent, suggesting the model tends to default to the dominant class when it is uncertain.

8.2. How Larger vs Smaller Models Fail Differently

To analyze how models of different sizes fail differently we focus on Experiment 11 (Section 6.3.10) as the reference experiment since it is the best performing experiment and contains all three model sizes with all four prompting strategies. The overall weighted F1 results are shown in Table 12 and the per-class breakdown for the few-shot strategy across all three models is shown in Table 16.

Model	Class	Precision	Recall	F1
Qwen3.5-0.8B	Ambivalent	0.59	0.51	0.55
Qwen3.5-0.8B	Clear Non-Reply	0.10	0.20	0.13
Qwen3.5-0.8B	Clear Reply	0.32	0.27	0.29
Qwen3.5-2B	Ambivalent	0.65	0.28	0.39
Qwen3.5-2B	Clear Non-Reply	0.23	0.44	0.30
Qwen3.5-2B	Clear Reply	0.36	0.64	0.46
Qwen3.5-4B	Ambivalent	0.55	0.10	0.16
Qwen3.5-4B	Clear Non-Reply	0.16	0.56	0.24
Qwen3.5-4B	Clear Reply	0.43	0.74	0.54

Table 16: Per-class results for few-shot strategy across all three models in Experiment 11 on the validation set

The three models show distinctly different behavior across the four prompting strategies:

- **Qwen3.5-0.8B** achieves the best overall result with few-shot (0.4274). The 0.8B model benefits most from few-shot because the Reason field explicitly explains why each example was assigned its label, helping the model better understand the classification logic behind each decision. This explicit reasoning guidance helps overcome its limited capacity by making the distinction between classes clearer. Zero-shot (0.3690) performs reasonably well as the second best strategy, suggesting the 0.8B model can follow simple instructions effectively without additional context. Persona (0.2958) performs worse than zero-shot, suggesting that the explicit three-step methodology adds complexity that the smallest model finds harder to follow consistently compared to the simple zero-shot instruction. CoT completely breaks down (0.1657) due to the high INVALID rate of 58.5% caused by the token limit of 150 tokens i.e. the 0.8B model generates reasoning chains that exceed the limit before producing the required Label: <label> line. The reason 0.8B outperforms the larger 2B model overall is that it better captures the dominant Ambivalent class (F1=0.55 vs 0.39 for 2B, as shown in Table 16), which has the highest weight in the weighted F1 score due to its high support (59.2% of the validation set).

- **Qwen3.5-2B** achieves its best result with few-shot (0.4037), closely followed by persona (0.3916) and zero-shot (0.3445). Similar to the 0.8B model, the 2B model benefits from few-shot because the Reason field helps it understand the classification logic behind each decision. However unlike the 0.8B model, persona also performs well (0.3916) suggesting that the 2B model has enough capacity to handle the complexity of the explicit three-step methodology effectively, which the 0.8B model struggles to follow consistently as it lacks the capacity to process the additional complexity. The reason 2B does not outperform 0.8B despite being a larger model is that it tends to under-predict the dominant Ambivalent class (recall=0.28, F1=0.39) and over-predict Clear Reply (recall=0.64), as shown in Table 16, which hurts its weighted F1 score since Ambivalent carries the highest weight. CoT completely breaks down (0.0679) with the highest INVALID rate of

92.5% across all three models, suggesting that the 2B model generates the longest reasoning chains which get cut off before reaching the required Label: <label> line.

- **Qwen3.5-4B** achieves its best result with zero-shot (0.3548), followed by persona (0.3438) and few-shot (0.2876). Unlike the 0.8B and 2B models, the 4B model performs best with the simplest strategy zero-shot. As shown in Table 16, the 4B model with few-shot almost never predicts Ambivalent (recall=0.10, F1=0.16). This suggests that rather than benefiting from additional guidance, the 4B model is actually harder to guide with labeled examples or explicit methodology, possibly because its larger size makes it more sensitive to the additional context provided. Adding labeled examples through few-shot actually hurts the 4B model significantly (dropping from 0.3548 to 0.2876), and persona also does not provide a significant improvement over zero-shot (0.3438 vs 0.3548). CoT breaks down (0.1176) with an INVALID rate of 87.2%, confirming that all models struggle with the token limit for CoT regardless of size. Despite being the largest model, the 4B model achieves the worst overall weighted F1 among all three models, confirming that model size alone is not the determining factor for performance on this task.

In summary the three models fail in fundamentally different ways: the 0.8B model defaults to the dominant Ambivalent class when uncertain but achieves the best overall balance across all strategies, the 2B model has more capacity to handle complex prompts but under-predicts Ambivalent and over-predicts Clear Reply, and the 4B model is harder to guide with additional context and performs best with the simplest strategy. These results confirm that model size alone is not the determining factor for performance on this task. The smallest model (0.8B) achieves the best overall weighted F1 by maintaining a more balanced prediction distribution that better reflects the true class distribution of the dataset. There appears to be an optimal match between model size and prompting complexity i.e. smaller models need more guidance through few-shot examples with explicit reasoning, while larger models perform better with simpler prompts and do not always benefit from additional context.

8.3. How Different Prompting Methods Fail Differently

To analyze how different prompting methods fail differently we focus on Experiment 11 (Section 6.3.10) and Experiment 13 (Section 6.3.12) as the reference experiments. Experiment 11 is the best performing experiment across all 13 experiments and Experiment 13 builds directly on top of it, investigating among other things the effect of increasing the CoT token limit to 400, providing a more complete picture of how each strategy behaves. Each strategy fails for a fundamentally different reason:

- **Zero-shot:** Zero-shot fails because without any examples or explicit methodology the model relies only on the label definitions in the system prompt. The definitions alone are not sufficient to distinguish the subtle boundaries between Ambivalent and Clear Reply, or between Ambivalent and Clear Non-Reply. As we saw the majority of both Clear Reply and Clear Non-Reply misclassifications are predicted as Ambivalent, confirming that the model defaults to the dominant class when uncertain. In Experiment 11 the zero-shot weighted F1 scores

were 0.3690 for Qwen3.5-0.8B, 0.3445 for Qwen3.5-2B and 0.3548 for Qwen3.5-4B. Zero-shot never reaches the performance of few-shot or persona across all experiments, confirming that the model needs more than just label definitions to classify political responses reliably.

- **Few-shot:** Few-shot is the best performing strategy overall, but it still fails on Clear Non-Reply and Clear Reply. With only one labeled example per class the model has limited exposure to the variety of ways each label can manifest. The Reason field added in Experiments 11 and 13 significantly improved performance (0.8B went from 0.3955 to 0.4274, 2B went from 0.3747 to 0.4037) by helping the model understand the classification logic. However even with the Reason field the model still struggles with Clear Non-Reply (F1=0.13 on validation) because one example is not enough to cover the full variety of refusal patterns.

- **CoT:** CoT fails for a completely different reason than the other strategies i.e. not primarily because of wrong classifications but because of INVALID predictions. In Experiment 11 with max_new_tokens=150 the INVALID rates were extremely high: 58.5% for Qwen3.5-0.8B, 92.5% for Qwen3.5-2B and 87.2% for Qwen3.5-4B. The models generated reasoning chains that exceeded the token limit causing the output to be cut off before reaching the required Label: <label> line. In Experiment 13 we increased max_new_tokens to 400 which dramatically reduced the INVALID rates to 18.0% for Qwen3.5-0.8B, 22.2% for Qwen3.5-2B and 23.0% for Qwen3.5-4B. However even with more valid predictions the CoT strategy still did not outperform the best strategy (Qwen3.5-0.8B + few-shot, 0.4247), the CoT weighted F1 scores in Experiment 13 were 0.2735 for Qwen3.5-0.8B (vs 0.4247 for Qwen3.5-0.8B few-shot), 0.2433 for Qwen3.5-2B (vs 0.3697 for Qwen3.5-2B few-shot) and 0.3534 for Qwen3.5-4B (vs 0.3161 for Qwen3.5-4B few-shot). This suggests that the step-by-step reasoning does not lead to better classifications than simply showing the model labeled examples with explicit reasoning in the Reason field.

- **Persona:** The original persona prompt (used in Experiments 1 to 10) failed because the role description focused on "evasion tactics" and "how politicians use language to avoid direct answers", which may have biased the model as confirmed by the per-class results in Table 2. The improved persona prompt (used in Experiments 11 and 13) addressed this bias as described in Section 4.4 (page 10). Comparing to Experiment 3, the improved persona significantly helped Qwen3.5-4B (from 0.2654 to 0.3438, a gain of 0.0784), had a negligible effect on Qwen3.5-2B (from 0.4042 to 0.3916, a drop of 0.0126 which is within the range of GPU non-determinism) and slightly hurt Qwen3.5-0.8B (from 0.3363 to 0.2958). However persona still does not beat few-shot for the 0.8B and 2B models, suggesting that concrete labeled examples with explicit reasoning are more effective than an expert role description with an explicit methodology alone.

In summary each prompting strategy fails for a different reason: zero-shot lacks concrete guidance, few-shot is limited by the number of examples, CoT is limited by the number of tokens and still does not improve over few-shot even when the token limit is increased, and persona is sensitive to the framing of the role description.

9. Implementation Notes

During the final submission in kaggle, the persona with few-shot hybrid and persona with CoT hybrid strategies were not run for Qwen3.5-4B. The persona with few-shot hybrid requires reducing BATCH_SIZE from 4 to 2 due to oom errors, which roughly doubles the inference time for this strategy. The persona with CoT hybrid uses CoT which is significantly slower than label-only strategies. The total inference time across all models and strategies already reached **9 hours and 19 minutes, and adding these two strategies for the 4B model would require several additional hours since the 4B model is by far the slowest of the three models and these two strategies are the most time consuming to run**. Since the 4B model did not achieve the best scores with any strategy in Experiments 11 and 13 and since the hybrid strategies did not outperform the simple few-shot strategy for the 0.8B and 2B models either, **running these additional combinations in the final kaggle submission was not considered worth the extra hours of inference time they would require**.

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